AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

Trainer: Aravindraj.G- Nminds Academy

AWS Academy Certified Educator

Create an EC2 Linux Instance in AWS

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Objective

EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

An EC2 Linux instance is a virtual machine running the Linux operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, Apache,FileServer etc.) in the cloud without having to manage physical hardware.

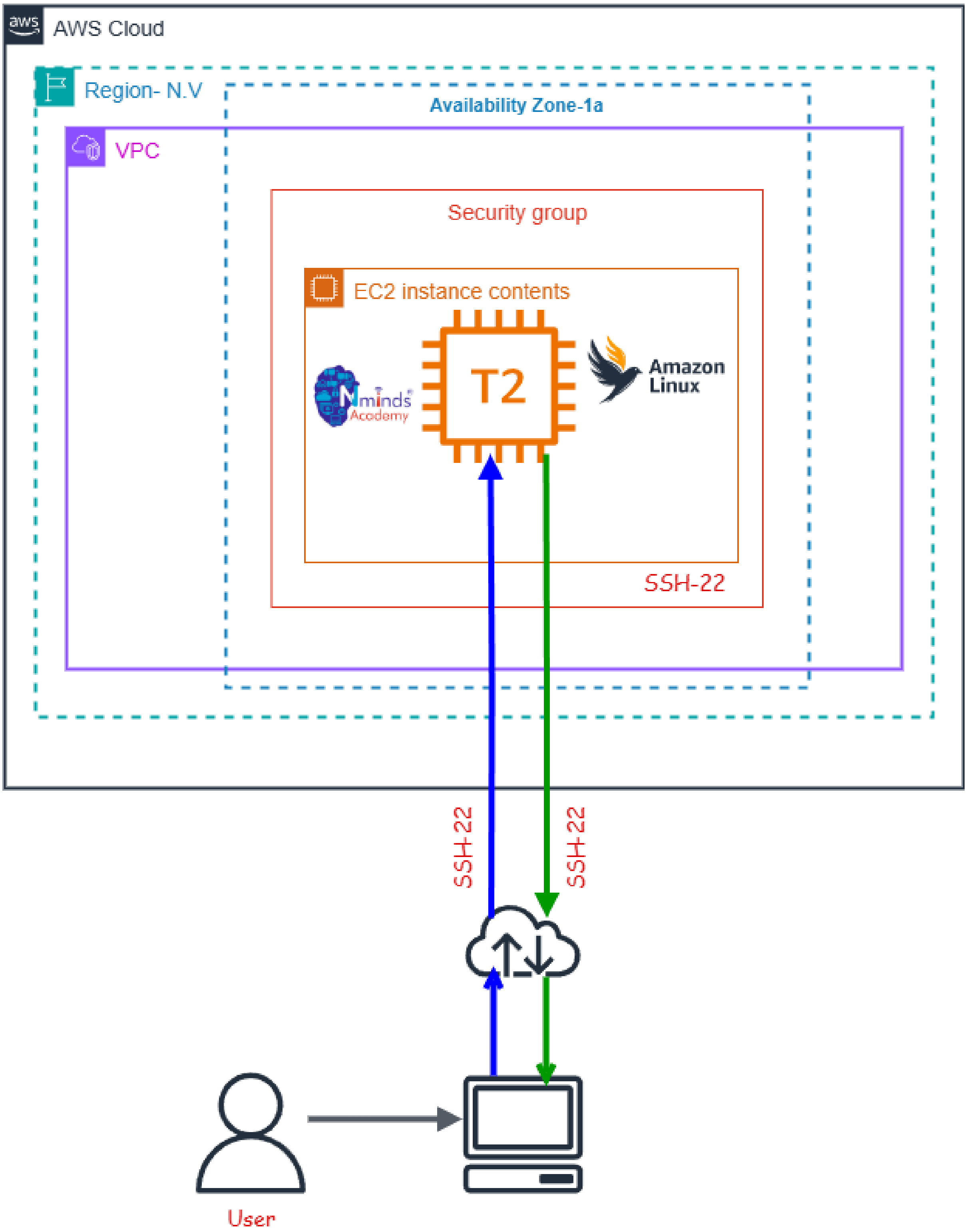
Use Cases for EC2 Linux Instances:

- Web and Application Hosting: Run Apache web servers or other Linux-based web applications in the cloud.
- Database Servers: Host SQL Server databases with EC2 Linux instances, using SSH or EC2 for more control.
- Remote Desktop Applications: Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.
- Development and Testing: Create development environments with linux OS to test php,MySql or other Linux-based applications.

Used Services in AWS

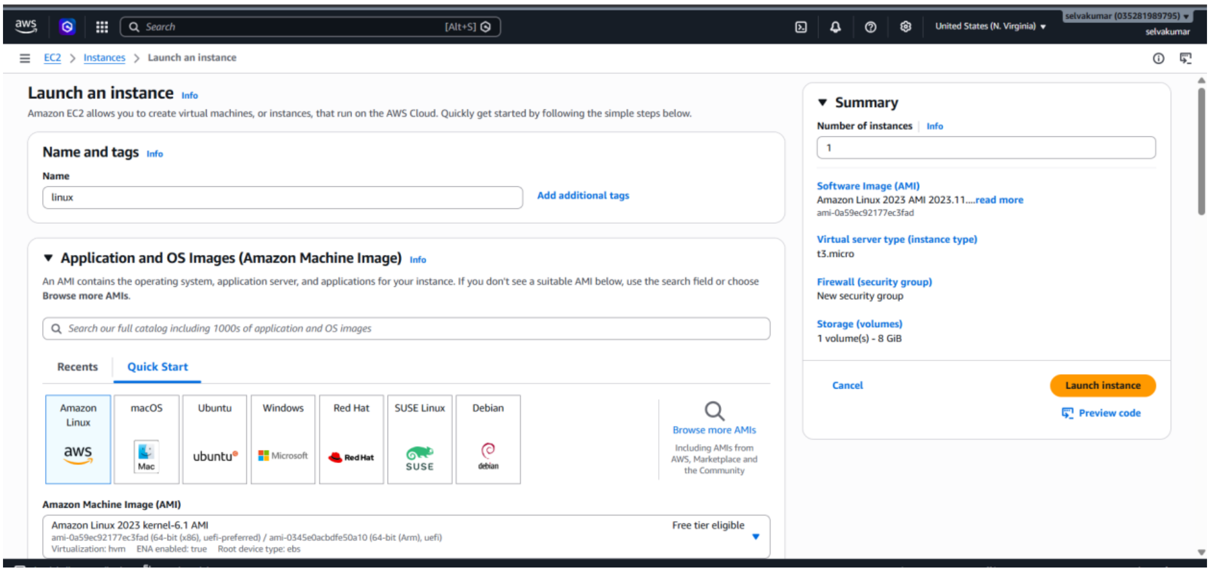
Amazon EC2

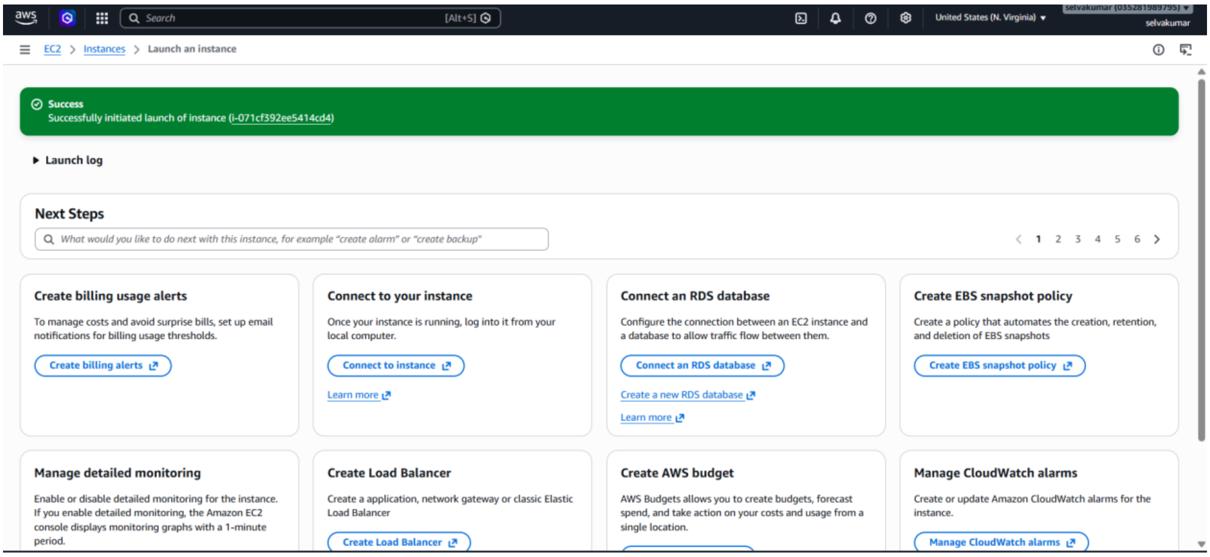
Topology

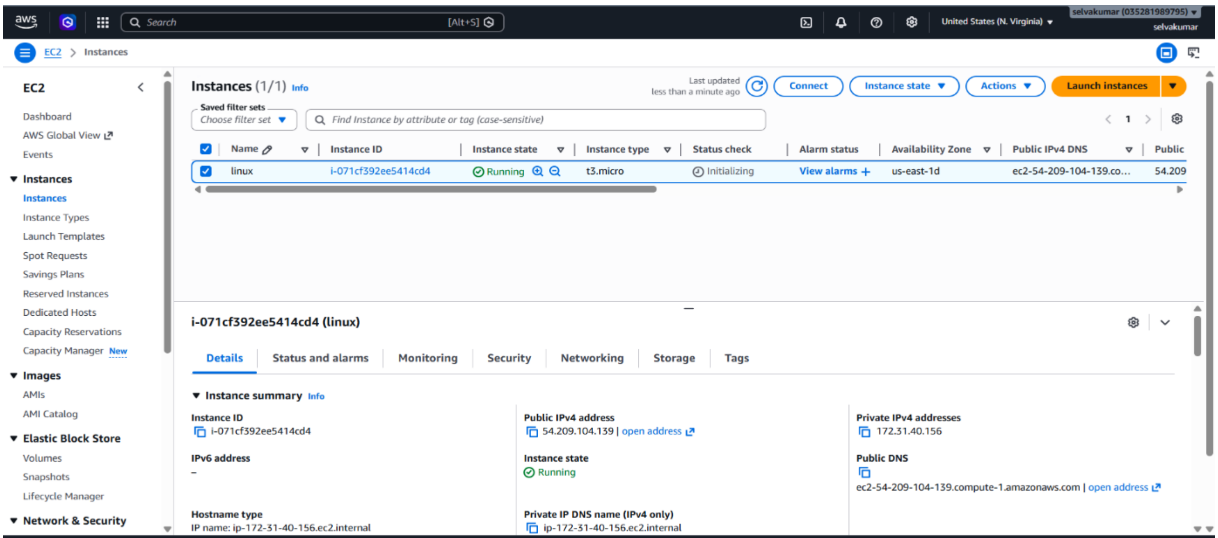


AWS Cloud
Region- N.V
Availability Zone-1a
VPC
Security group
EC2 instance contents
T2
Amazon Linux
SSH-22
SSH-22
SSH-22
User

Execution Tasks:







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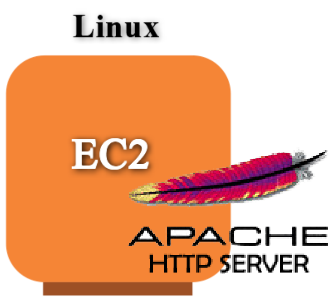
Configure Apache Web Server on Linux Instance in AWS



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Used Services in AWS

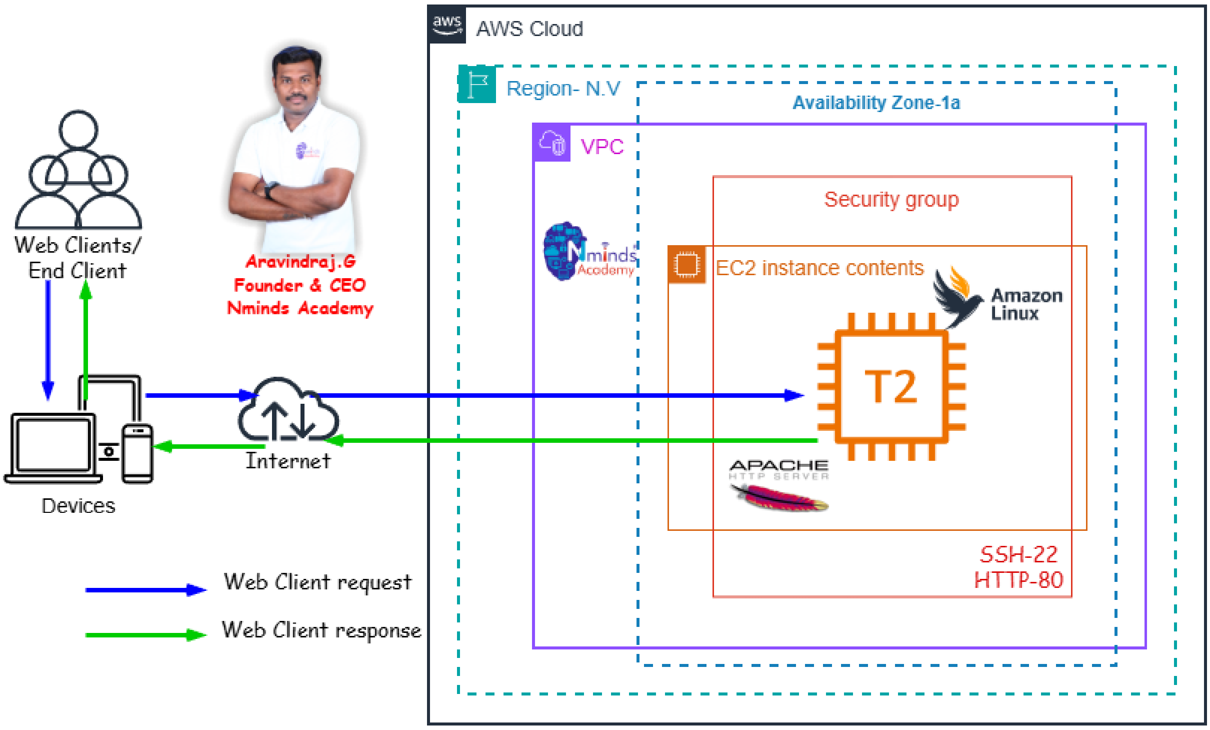
Amazon EC2

Used Services in Linux Server

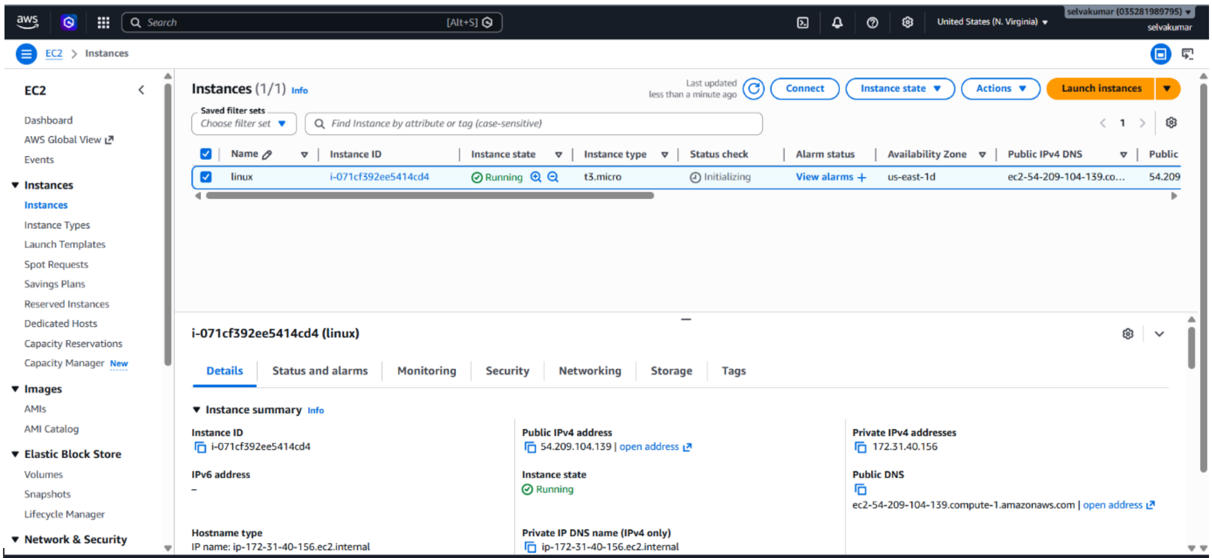
Apache Webserver

Topology

How to Configure the Apache web server in Linux Server with AWS EC2

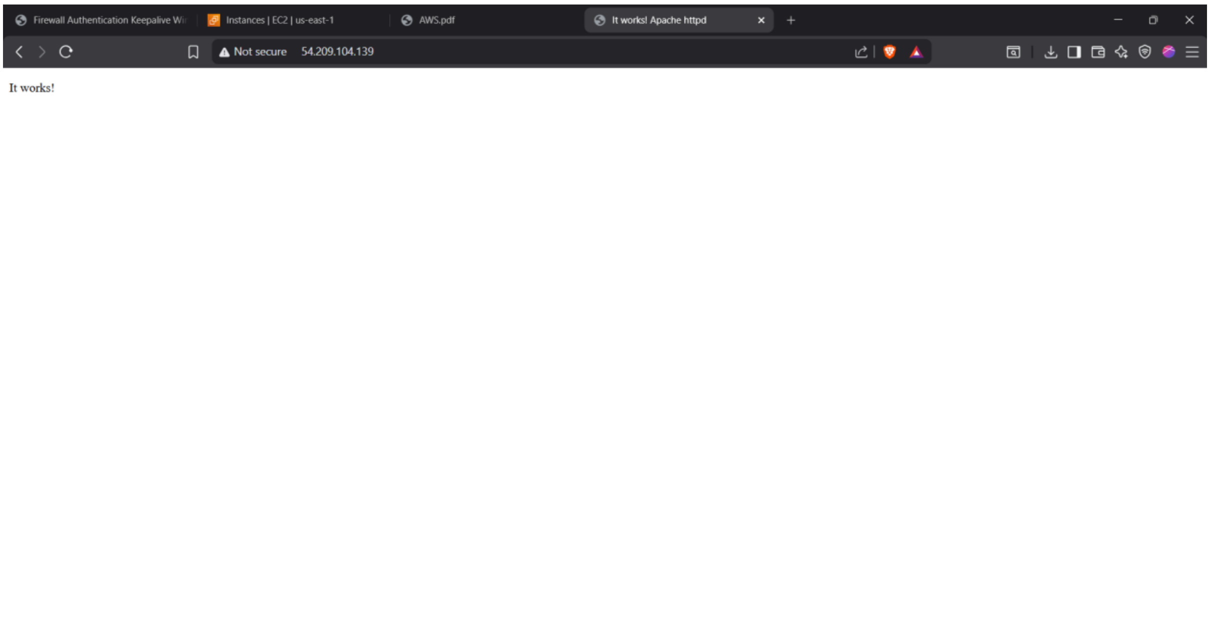


Execution Tasks:




```
[ec2-user@ip-172-31-40-156 ~]$ sudo -i
[root@ip-172-31-40-156 ~]# yum update -y
Amazon Linux 2023 Kernel Livepatch repository                              273 kB/s |  38 kB     00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-40-156 ~]# yum install httpd -y
Last metadata expiration check: 0:00:09 ago on Mon May 11 17:00:18 2026.
Dependencies resolved.
========================================================================================================
 Package                    Architecture     Version                        Repository            Size
========================================================================================================
Installing:
 httpd                      x86_64           2.4.66-1.amzn2023.0.1          amazonlinux           47 k
Installing dependencies:
 apr                        x86_64           1.7.5-1.amzn2023.0.4           amazonlinux          129 k
 apr-util                   x86_64           1.6.3-1.amzn2023.0.2           amazonlinux           97 k
 apr-util-lmdb              x86_64           1.6.3-1.amzn2023.0.2           amazonlinux           13 k
 generic-logos-httpd        noarch           18.0.0-12.amzn2023.0.3         amazonlinux           19 k
 httpd-core                 x86_64           2.4.66-1.amzn2023.0.1          amazonlinux          1.4 M
 httpd-filesystem           noarch           2.4.66-1.amzn2023.0.1          amazonlinux           13 k
 httpd-tools                x86_64           2.4.66-1.amzn2023.0.1          amazonlinux           81 k
 libbrotli                  x86_64           1.0.9-4.amzn2023.0.2           amazonlinux          315 k
 mailcap                    noarch           2.1.49-3.amzn2023.0.3          amazonlinux           33 k
Installing weak dependencies:
 apr-util-openssl           x86_64           1.6.3-1.amzn2023.0.2           amazonlinux           15 k
 mod_http2                  x86_64           2.0.27-1.amzn2023.0.3          amazonlinux          166 k
 mod_lua                    x86_64           2.4.66-1.amzn2023.0.1          amazonlinux           60 k

Transaction Summary
========================================================================================================
Install  13 Packages

Total download size: 2.4 M
Installed size: 6.9 M
Downloading Packages:
(1/13): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm                                3.7 MB/s | 129 kB     00:00
(2/13): apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64.rpm                      360 kB/s |  13 kB     00:00
(3/13): apr-util-1.6.3-1.amzn2023.0.2.x86_64.rpm                           2.4 MB/s |  97 kB     00:00
(4/13): apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64.rpm                   659 kB/s |  15 kB     00:00
(5/13): httpd-2.4.66-1.amzn2023.0.1.x86_64.rpm                             2.0 MB/s |  47 kB     00:00
```

```
root@ip-172-31-40-156:~
  Verifying        : httpd-2.4.66-1.amzn2023.0.1.x86_64                                          6/13
  Verifying        : httpd-core-2.4.66-1.amzn2023.0.1.x86_64                                     7/13
  Verifying        : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch                               8/13
  Verifying        : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64                                    9/13
  Verifying        : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                                      10/13
  Verifying        : mailcap-2.1.49-3.amzn2023.0.3.noarch                                       11/13
  Verifying        : mod_http2-2.0.27-1.amzn2023.0.3.x86_64                                     12/13
  Verifying        : mod_lua-2.4.66-1.amzn2023.0.1.x86_64                                       13/13

Installed:
  apr-1.7.5-1.amzn2023.0.4.x86_64                         apr-util-1.6.3-1.amzn2023.0.2.x86_64
  apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64               apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch       httpd-2.4.66-1.amzn2023.0.1.x86_64
  httpd-core-2.4.66-1.amzn2023.0.1.x86_64                 httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch
  httpd-tools-2.4.66-1.amzn2023.0.1.x86_64                libbrotli-1.0.9-4.amzn2023.0.2.x86_64
  mailcap-2.1.49-3.amzn2023.0.3.noarch                    mod_http2-2.0.27-1.amzn2023.0.3.x86_64
  mod_lua-2.4.66-1.amzn2023.0.1.x86_64

Complete!
[root@ip-172-31-40-156 ~]# systemctl start httpd
[root@ip-172-31-40-156 ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
     Active: active (running) since Mon 2026-05-11 17:01:30 UTC; 9s ago
       Docs: man:httpd.service(8)
   Main PID: 26237 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 1014)
     Memory: 13.5M
        CPU: 92ms
     CGroup: /system.slice/httpd.service
             ├─26237 /usr/sbin/httpd -DFOREGROUND
             ├─26262 /usr/sbin/httpd -DFOREGROUND
             ├─26263 /usr/sbin/httpd -DFOREGROUND
             ├─26287 /usr/sbin/httpd -DFOREGROUND
             └─26299 /usr/sbin/httpd -DFOREGROUND

May 11 17:01:30 ip-172-31-40-156.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
May 11 17:01:30 ip-172-31-40-156.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
May 11 17:01:30 ip-172-31-40-156.ec2.internal httpd[26237]: Server configured, listening on: port 80
[root@ip-172-31-40-156 ~]#
```


Firewall Authentication Keepalive Wi
Instances | EC2 | us-east-1
AWS.pdf
It works! Apache httpd
Not secure
54.209.104.139
It works!